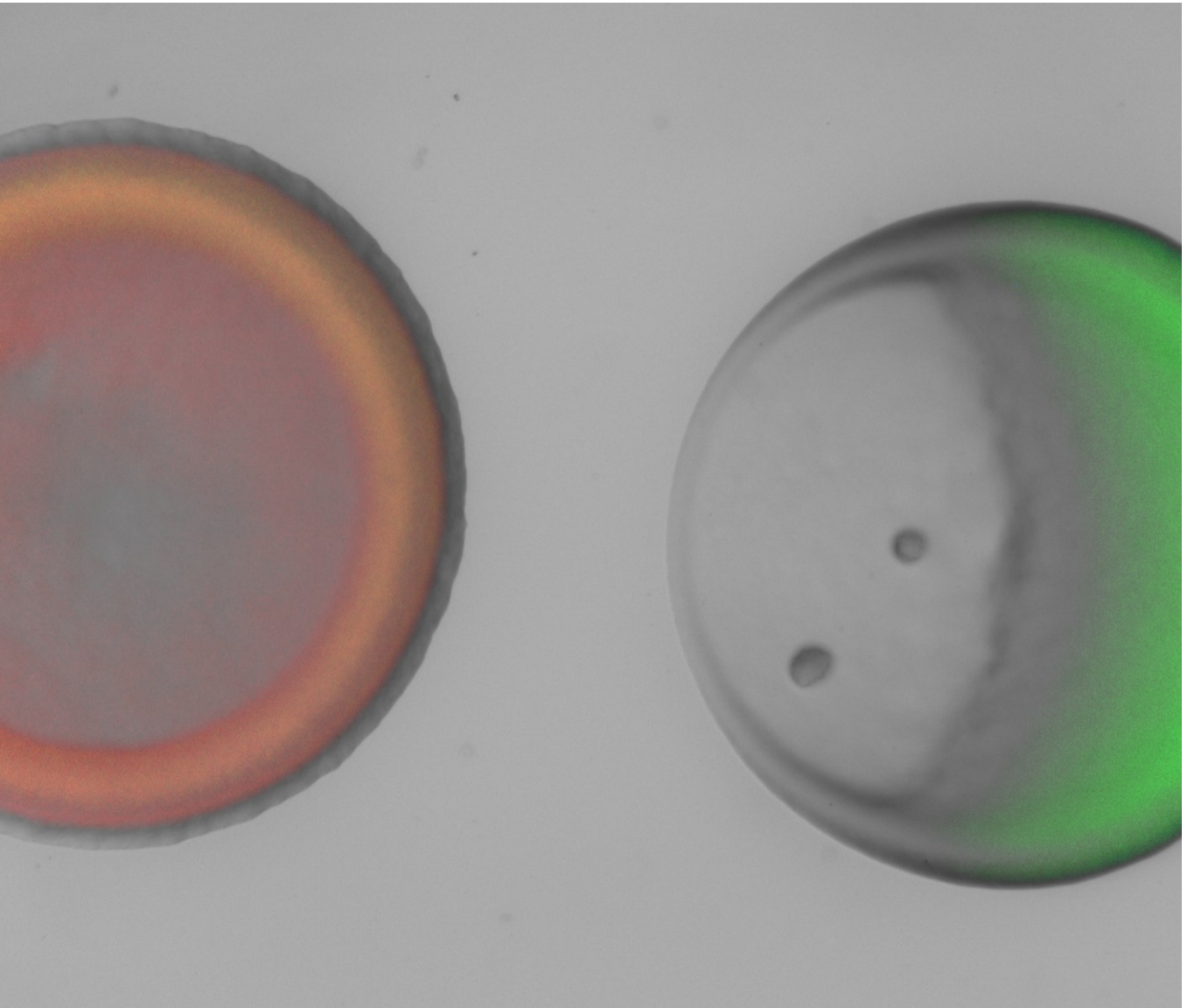
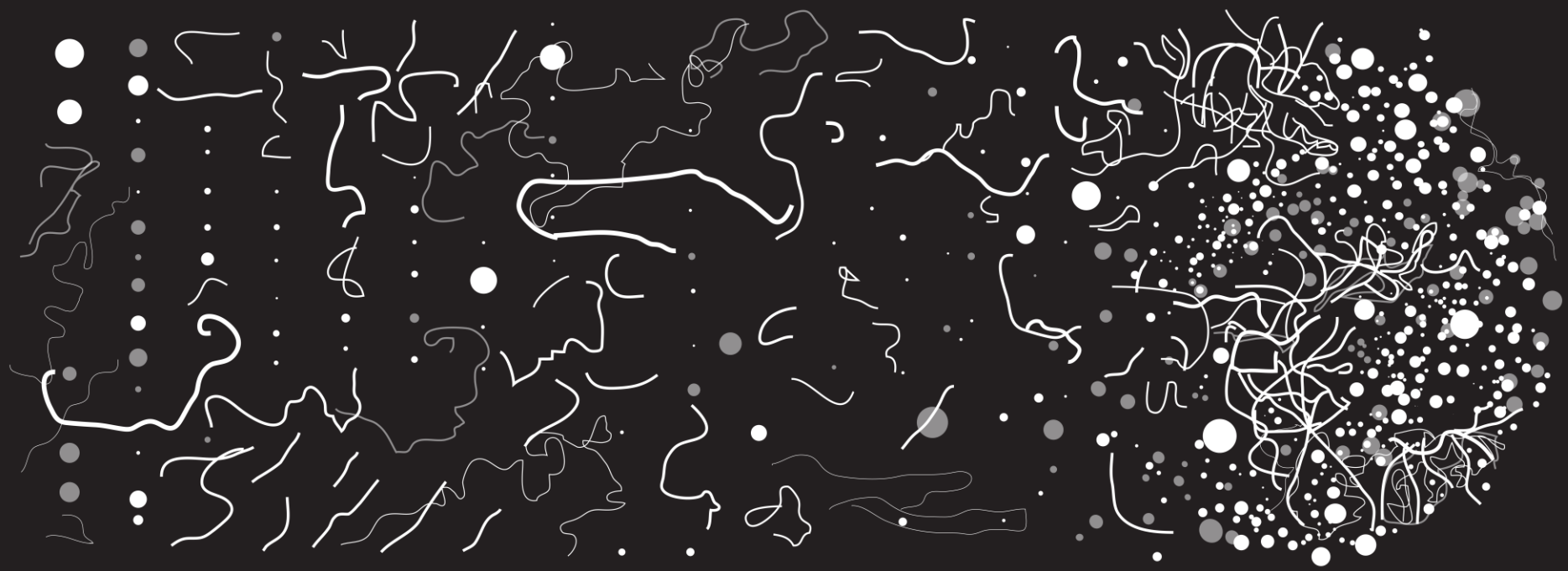


Computational design of synthetic microbial communities

Alex J H Fedorec
Division of Biosciences
University College London

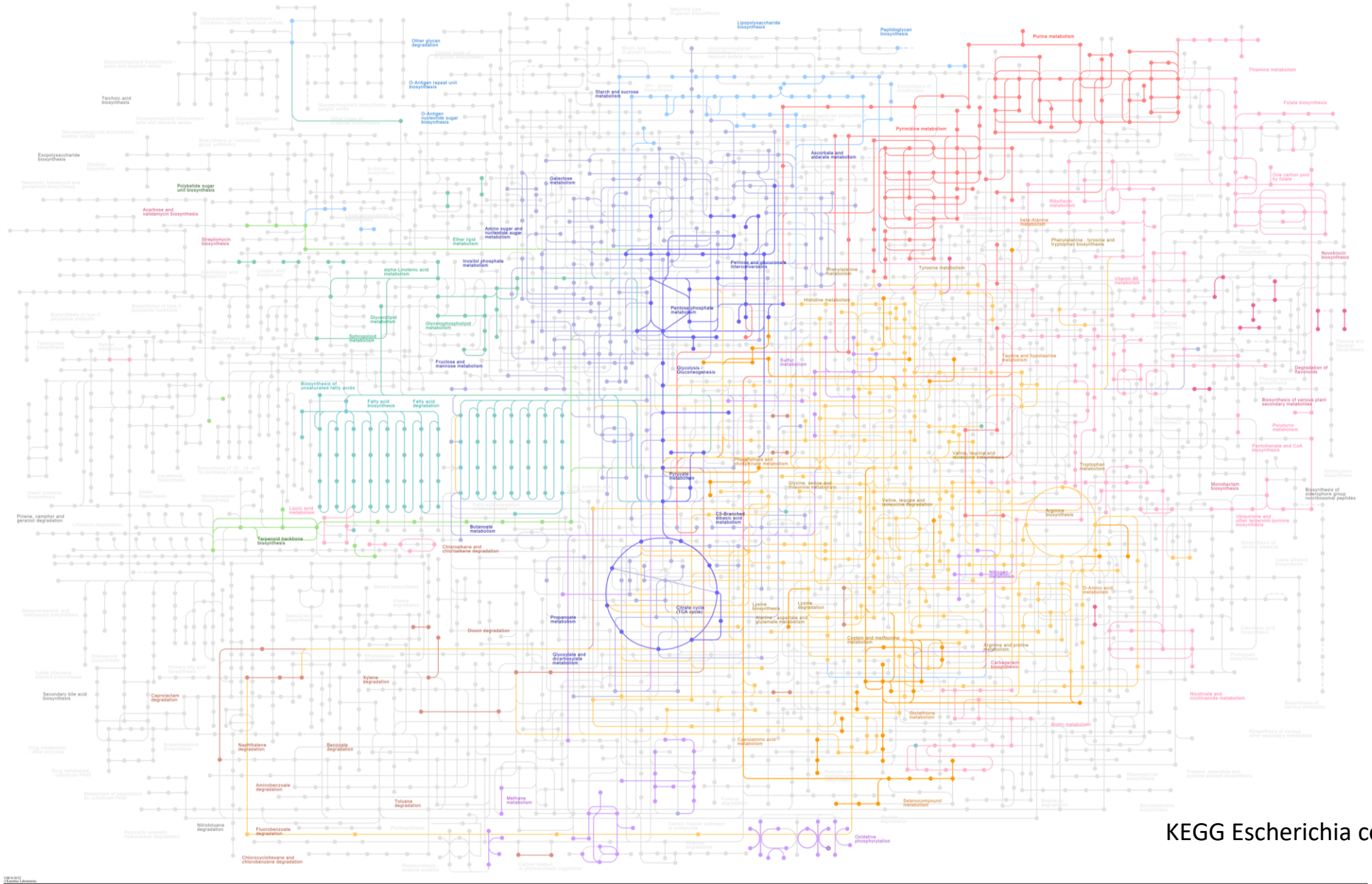
Why am I interested in microbial communities?



Sakurako Gibo

Microbes are (nearly) everywhere

That makes them a great tool for engineering



KEGG Escherichia coli K-12 MG1655



But it means there will be competition wherever we go

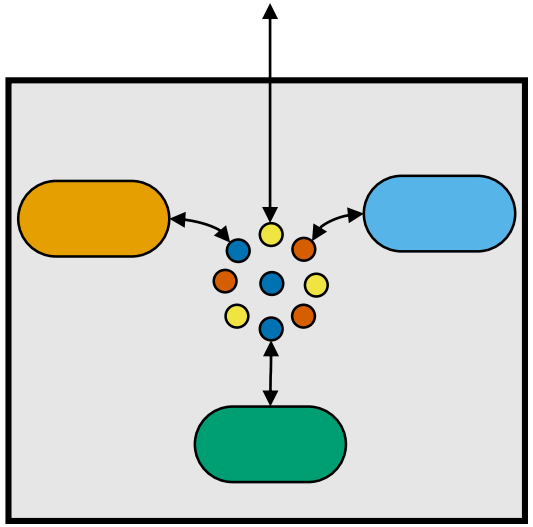


Even in a sterile environment, microbes have limitations

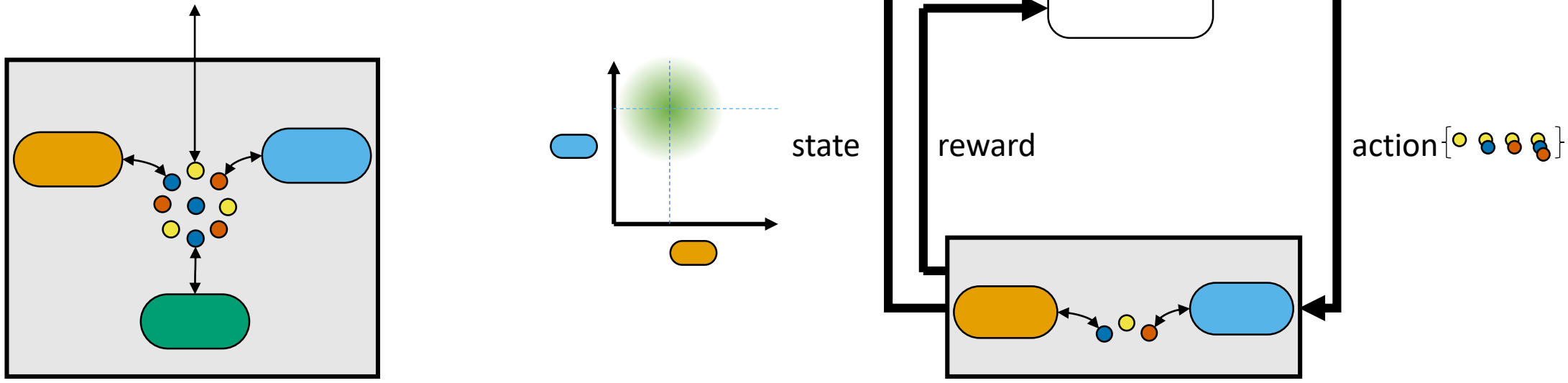

Therefore, creating synthetic communities *might* be useful

Synthetic microbial ecology

Cell-environment interactions

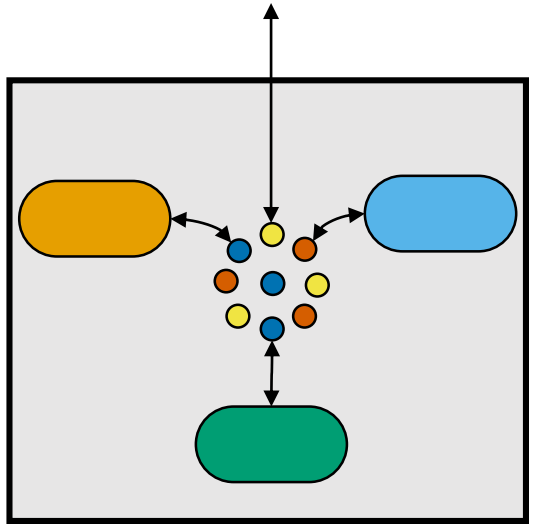


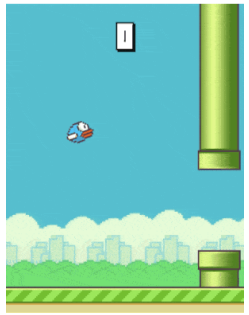
Synthetic microbial ecology – RL control



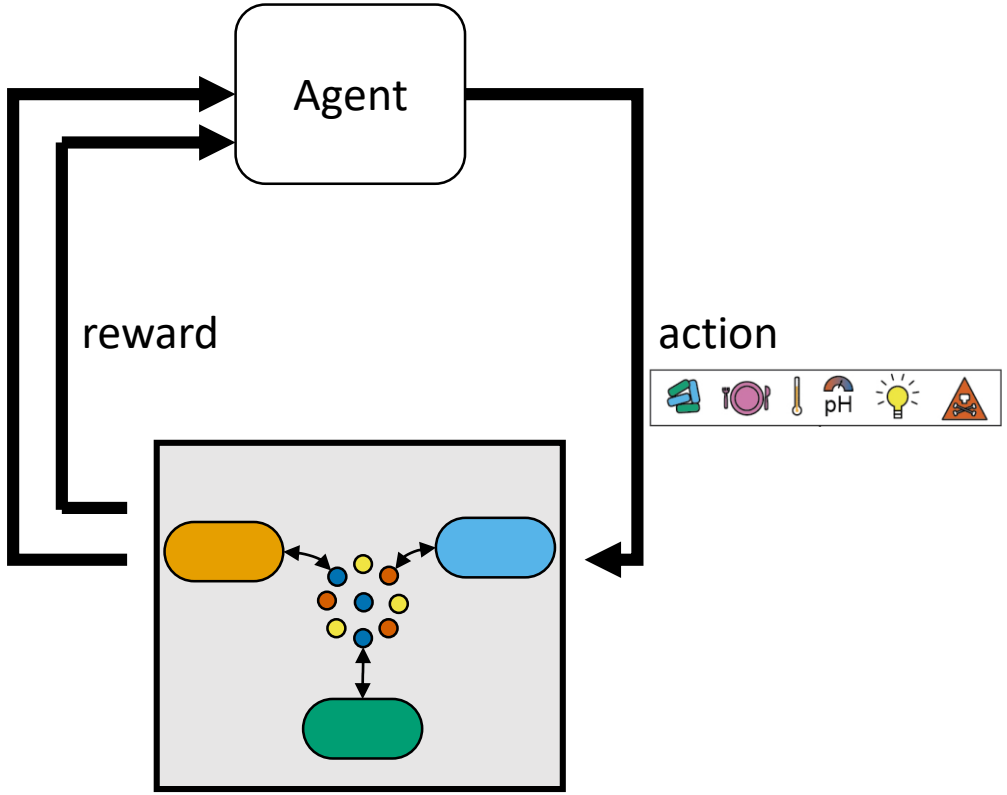
Treloar et al., PLOS Comp Biol, 2020

Synthetic microbial ecology – RL control



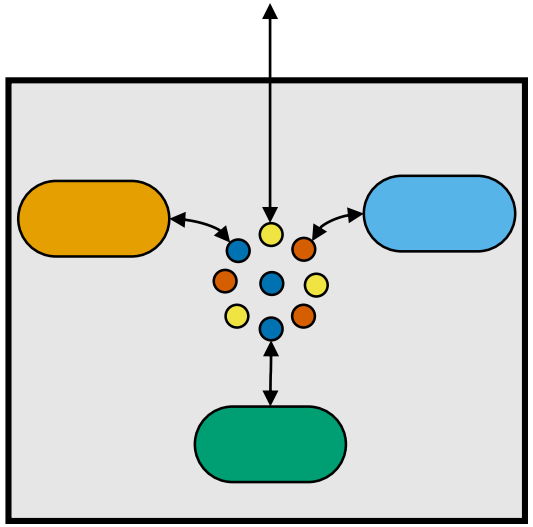


state

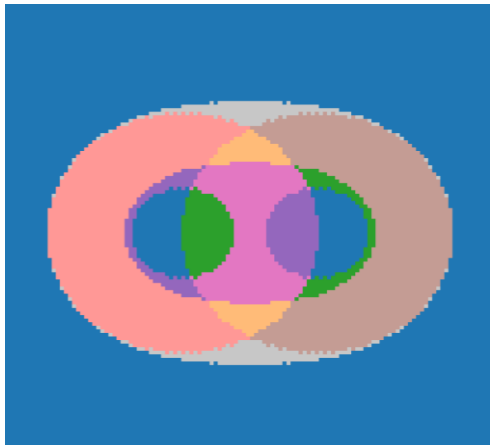


Synthetic microbial ecology

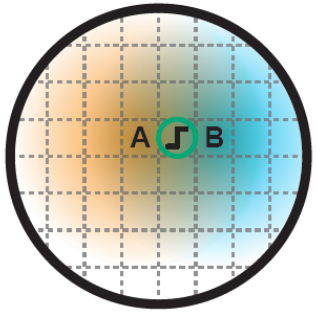
Cell-environment interactions



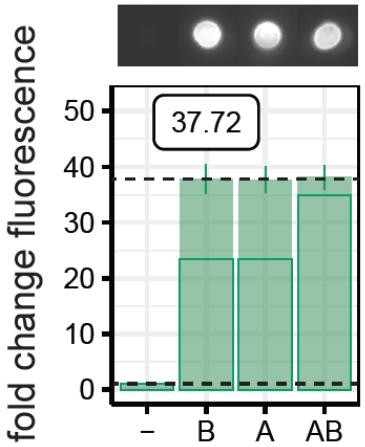
Spatial organisation

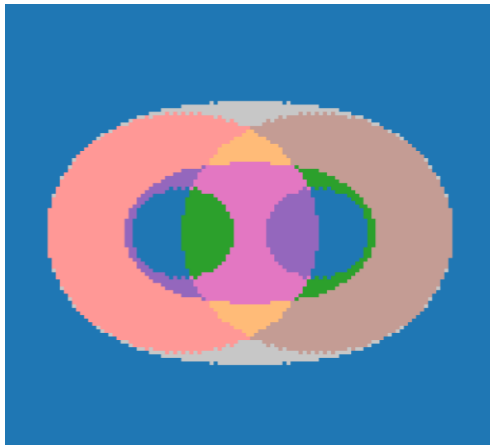


Synthetic microbial ecology – computational biofilms



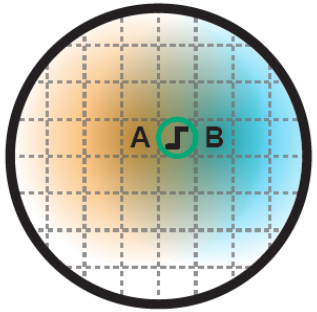
A	0	0	1	1
B	0	1	0	1
OR	0	1	1	1



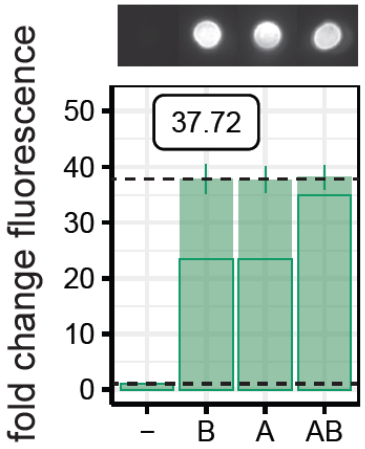


Fedorec et al., Nat Commun, 2024

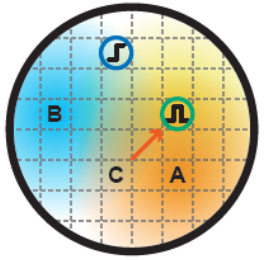
Synthetic microbial ecology – computational biofilms



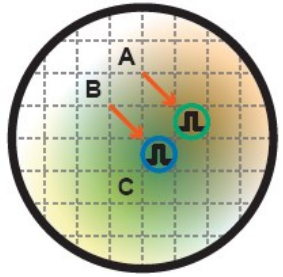
A	0	0	1	1
B	0	1	0	1
OR	0	1	1	1

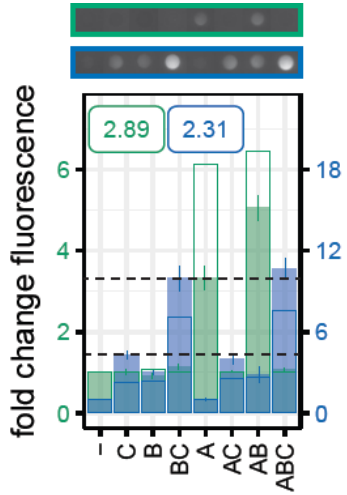


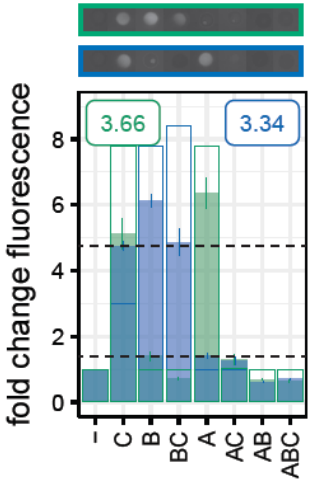
A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
(A > C) 0x0A	0	0	0	0	1	0	1	0
(B AND C) 0x11	0	0	0	1	0	0	0	1
(2-to-1 MUX) 0x1B	0	0	0	1	1	0	1	1

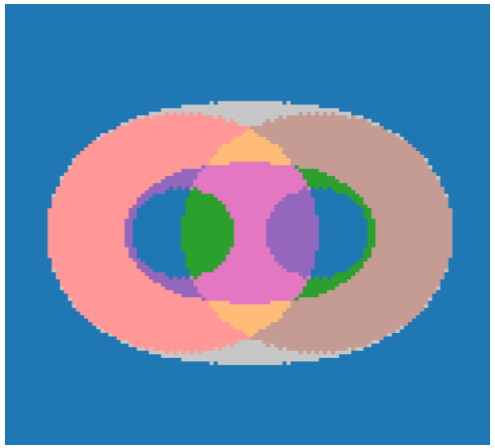


A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
0x70	0	1	1	1	0	0	0	0
0x48	0	1	0	0	1	0	0	0
(Rule 30) 0x78	0	1	1	1	1	0	0	0





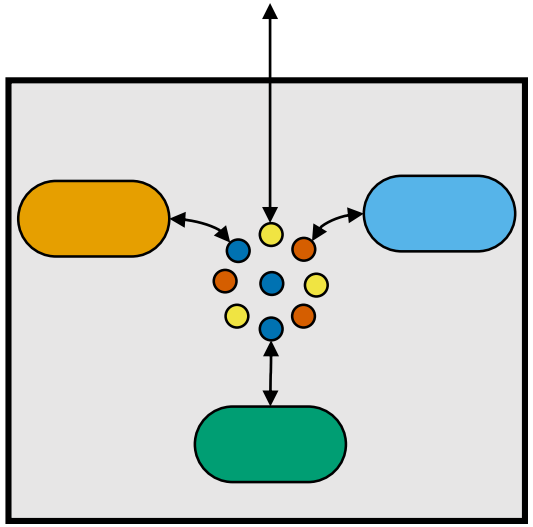




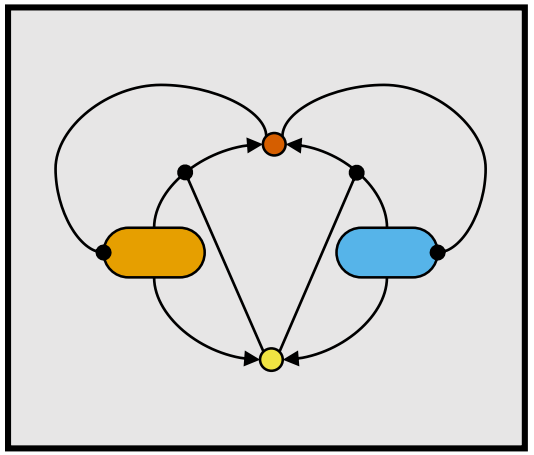
Fedorec et al., Nat Commun, 2024

Synthetic microbial ecology

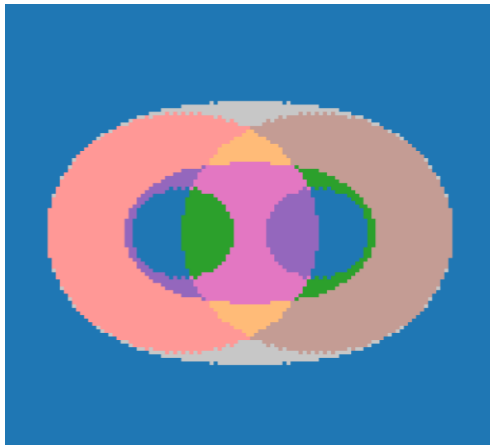
Cell-environment interactions



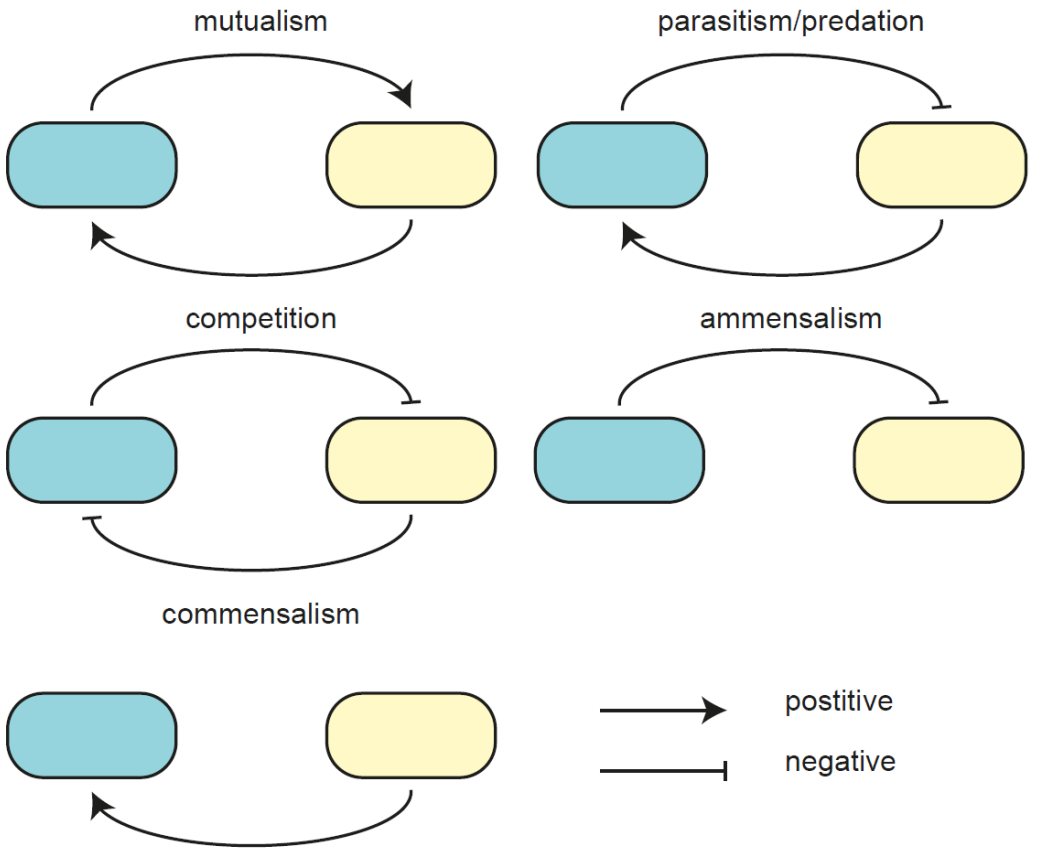
Cell-cell interactions



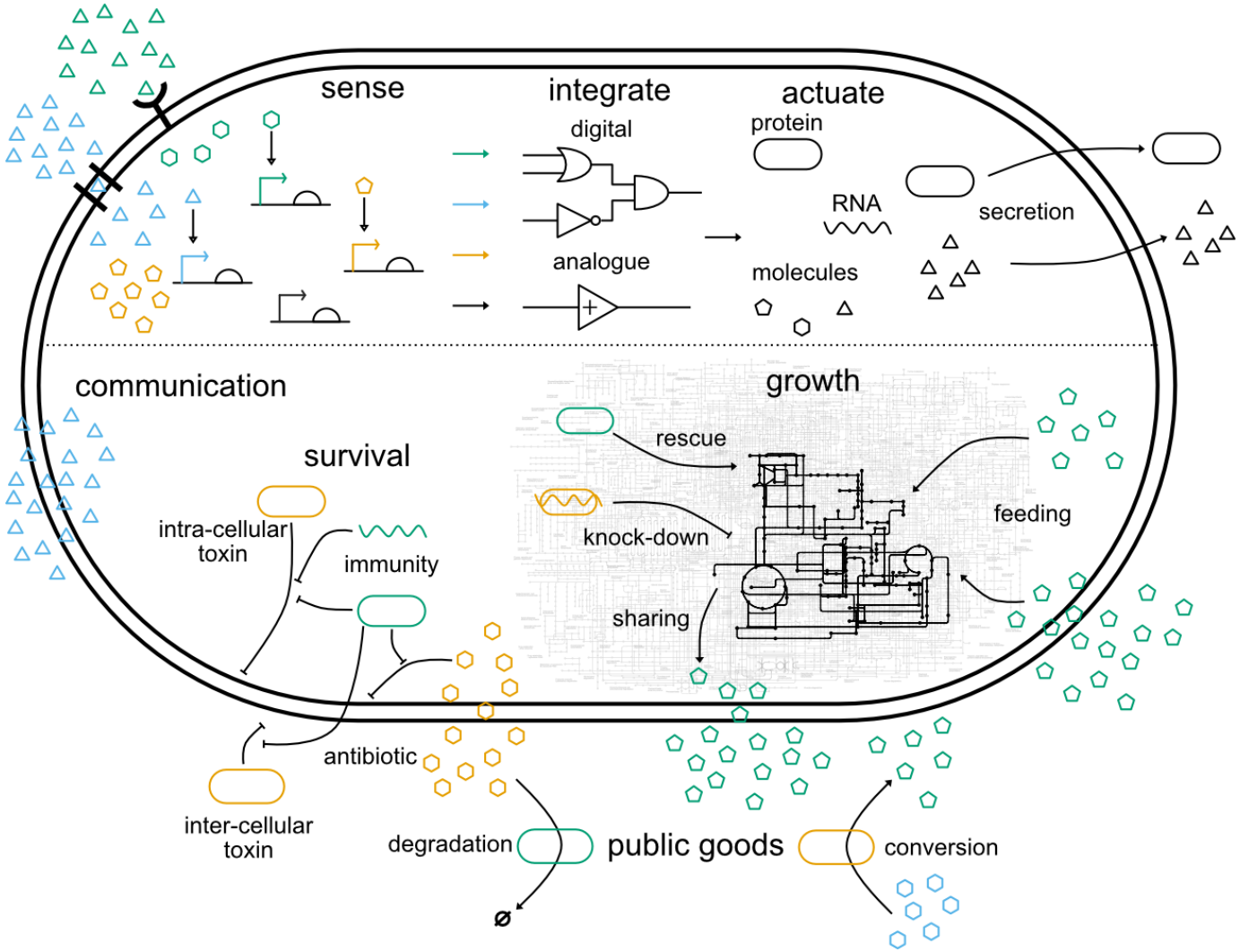
Spatial organisation/ computation



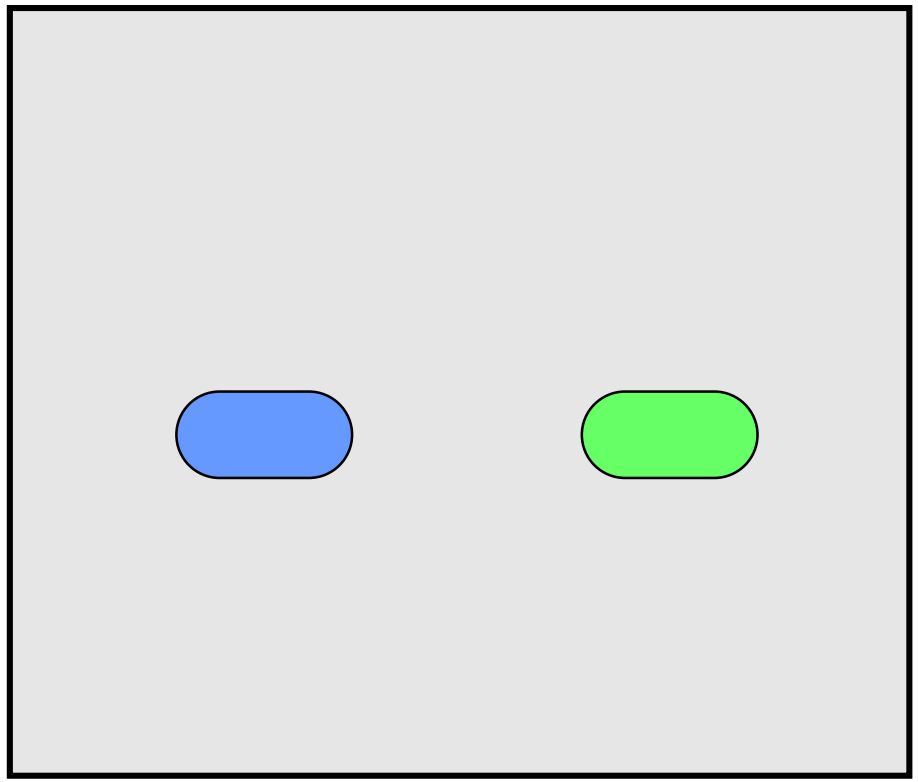
Interactions shape communities



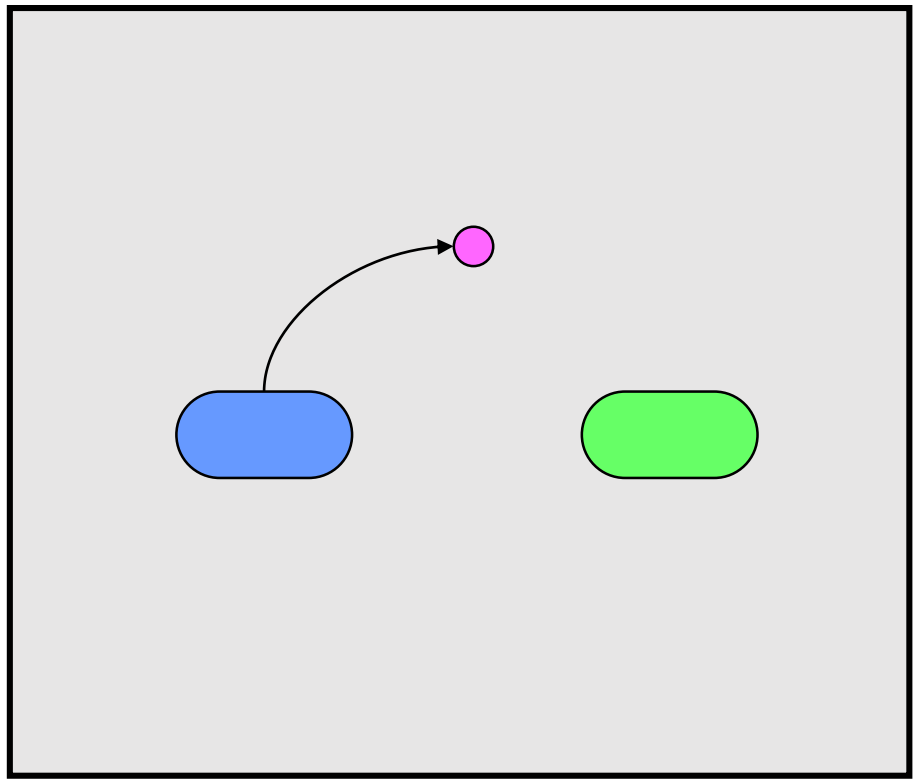
Synthetic biology allows us to engineer interactions



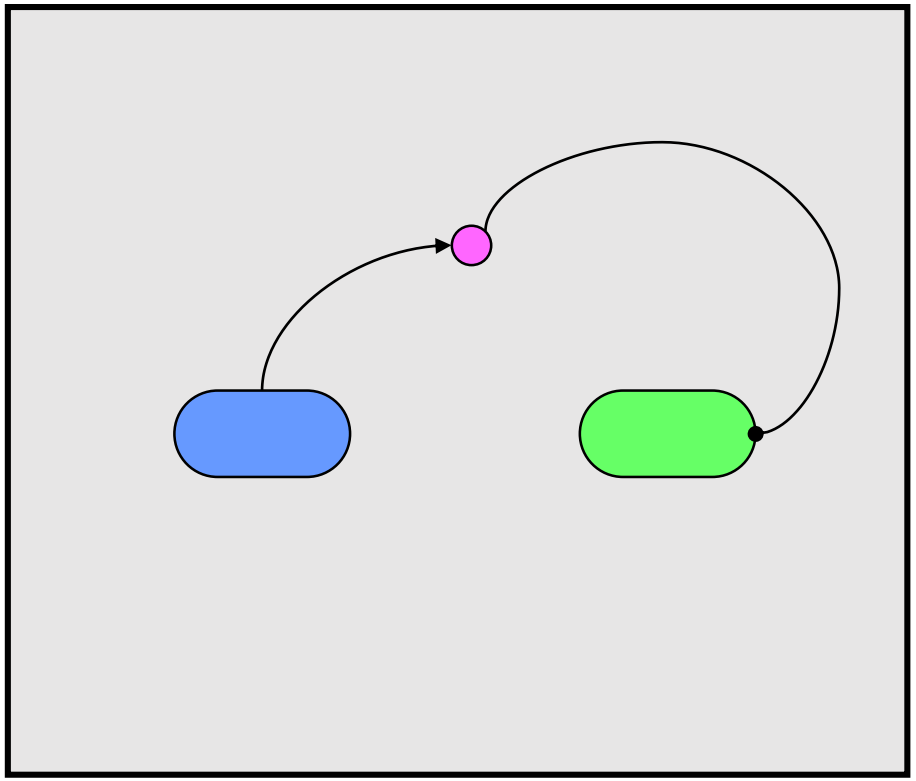
We can build new communities by engineering interactions



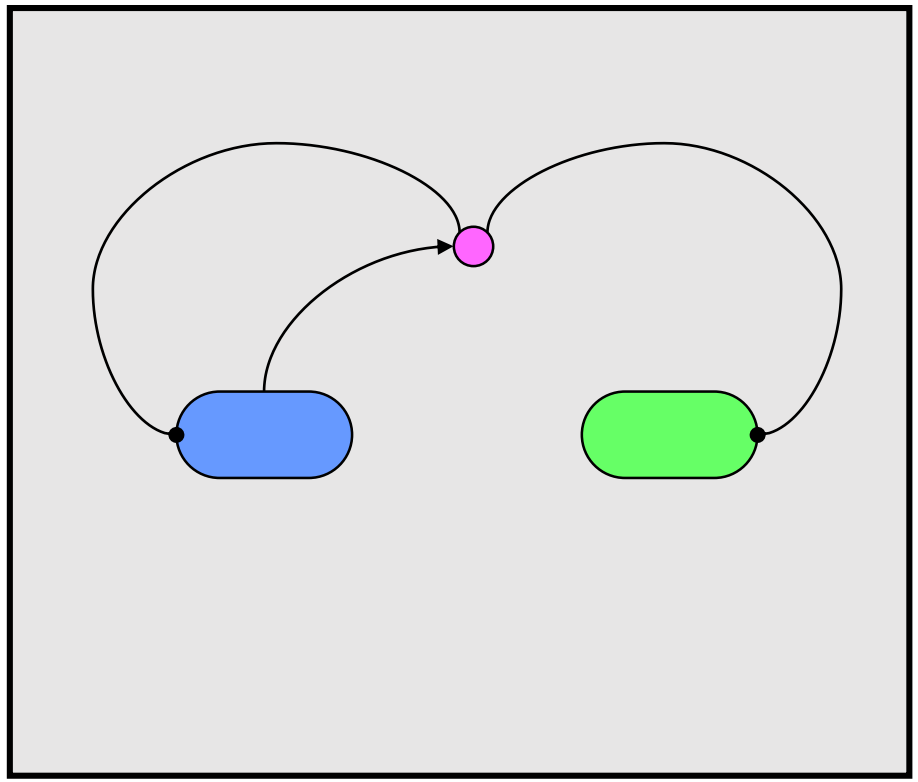
We can build new communities by engineering interactions



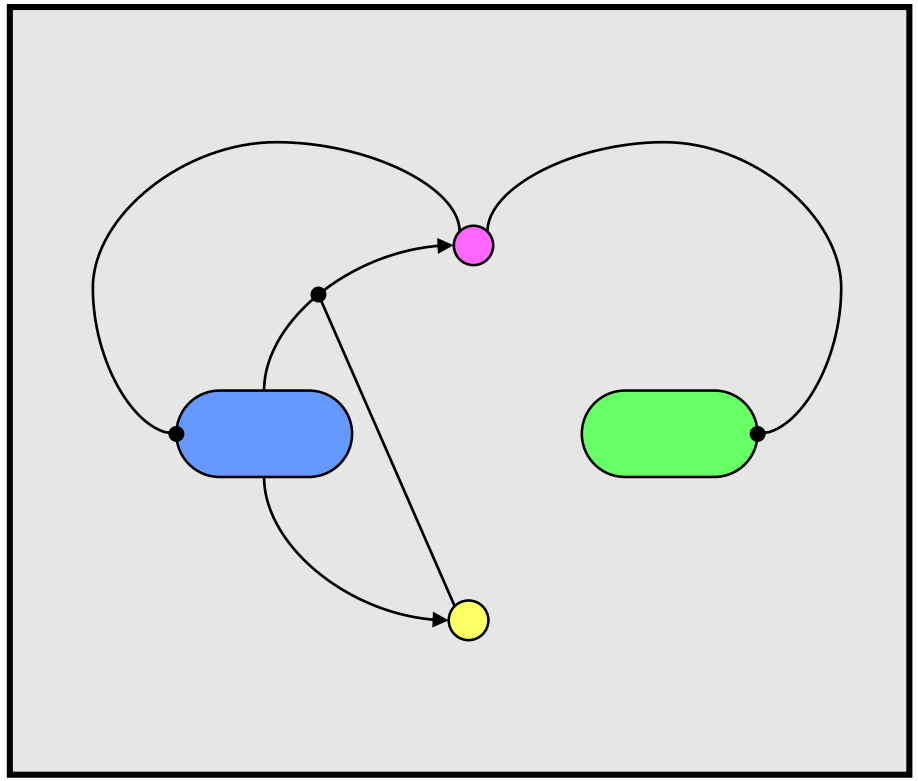
We can build new communities by engineering interactions



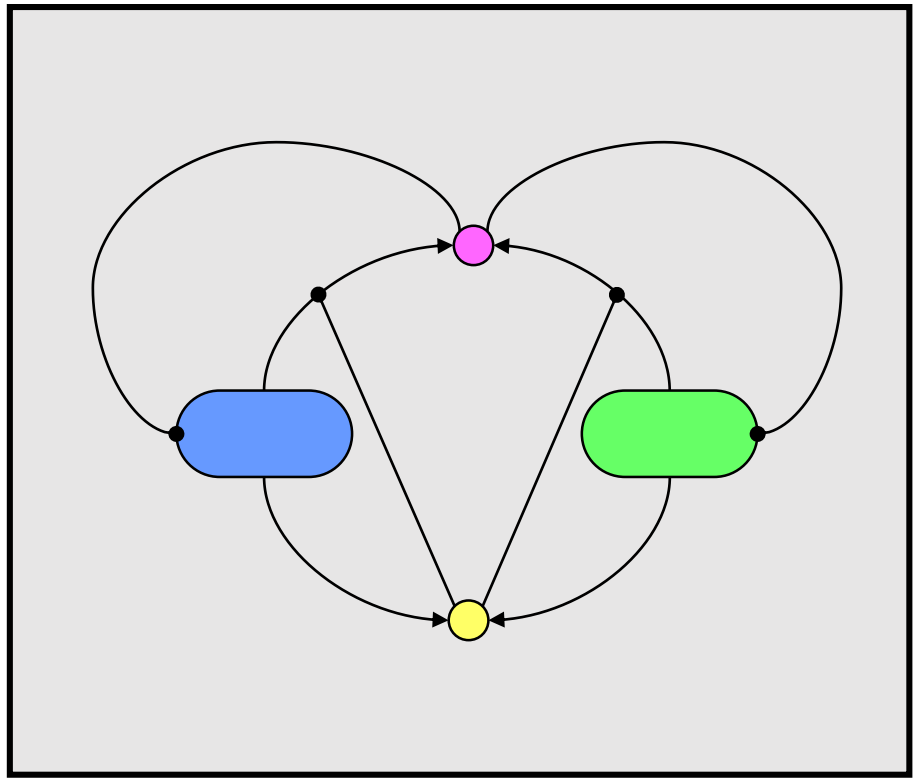
We can build new communities by engineering interactions



We can build new communities by engineering interactions



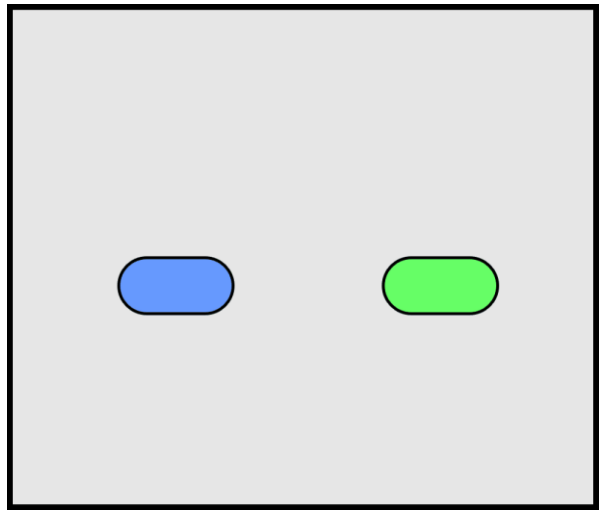
We can build new communities by engineering interactions



How do we know what to build?

For: 2 strains
1 extra-cellular molecule
1 signal molecule

15 unique topologies



How do we know what to build?

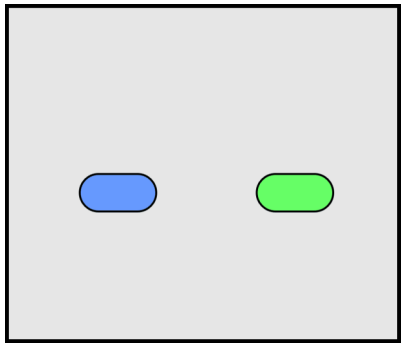
For: 2 strains
1 extra-cellular molecule
1 signal molecule

15 unique topologies

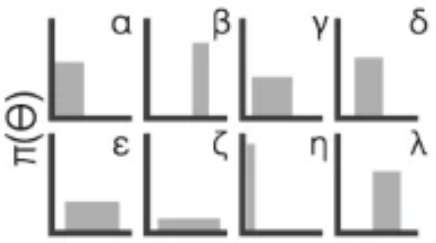
$$
\begin{aligned}
\frac{dX_i}{dt} &= \mu_i X_i - \omega_i X_i - D X_i \\
\frac{dA_j}{dt} &= D(A_{j_0} - A_j) + \sum_i \rho_{ij} X_i - \sum_i \frac{\mu_i X_i}{\gamma_{A_{ij}}} \\
\frac{dT_k}{dt} &= D(T_{k_0} - T_k) + \sum_i \rho_{ik} X_i - \sum_i \frac{\omega_i X_i}{\gamma_{T_{ik}}} \\
\frac{dS_m}{dt} &= D(S_{m_0} - S_m) + \sum_i k_{S_{im}} X_i \\
\mu_i &= \mu_{\max_i} \prod_j \frac{A_j}{K_{A_{ij}} + A_j} \\
\omega_i &= \sum_k \omega_{\max_{ik}} \frac{T_k}{K_{T_{ik}} + T_k} \\
\rho_{iz} &= \sum_m \left((k_{\max_{izm}} - k_{\min_{izm}}) \frac{S_m}{K_{izm} + S_m} + k_{\min_{izm}} \right)
\end{aligned}
$$

Inputs

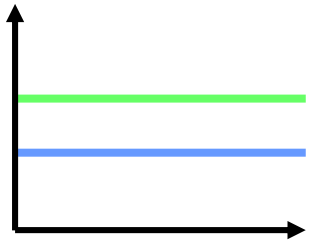
Model space



Parameter space



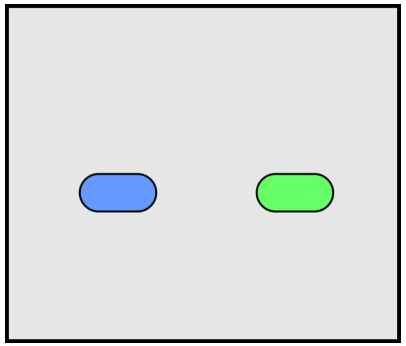
Desired behaviour



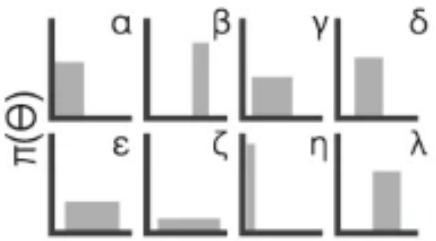
Karkaria et al., Nat Commun, 2021
Karkaria et al., PLOS Comp Biol, 2022

Inputs

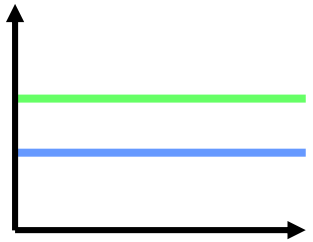
Model space



Parameter space

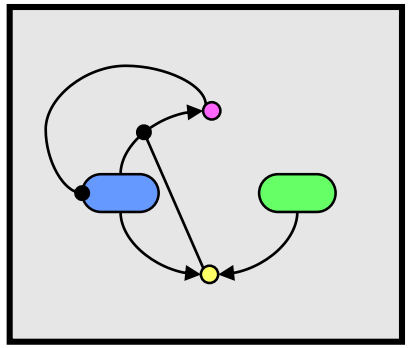


Desired behaviour

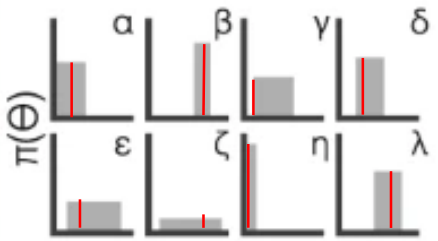


ABC

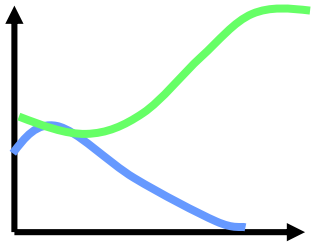
Sample a model



Sample parameters



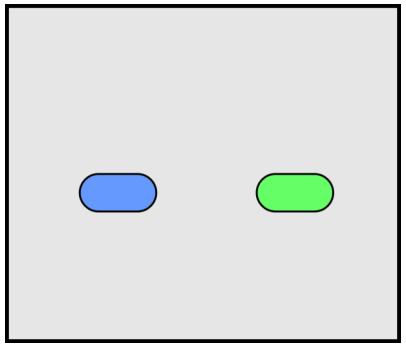
Simulate model and compare to desired behaviour



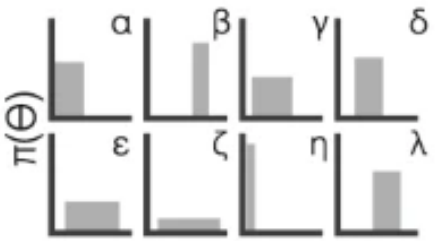
Karkaria et al., Nat Commun, 2021
Karkaria et al., PLOS Comp Biol, 2022

Inputs

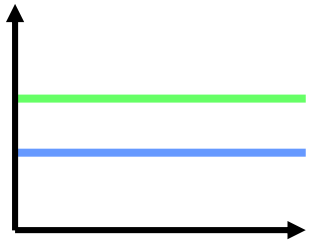
Model space



Parameter space

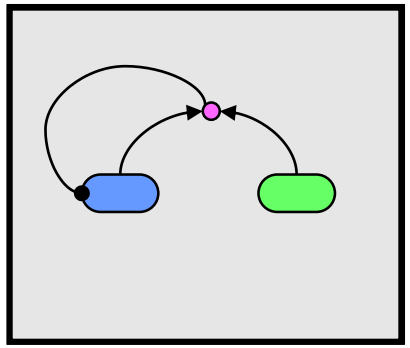


Desired behaviour

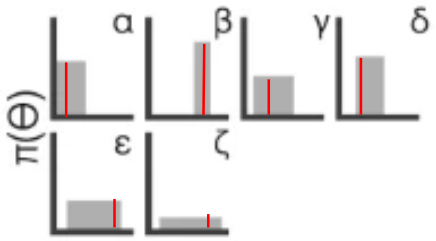


ABC

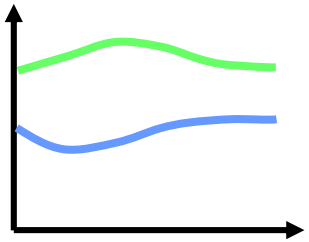
Sample a model



Sample parameters



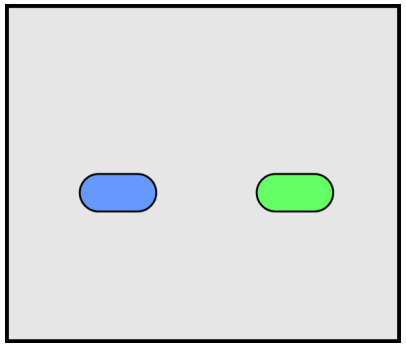
Simulate model and compare to desired behaviour



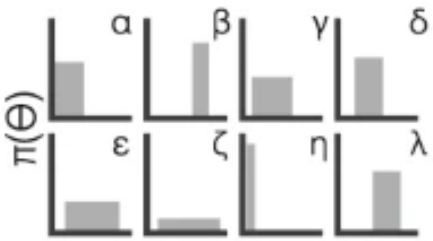
Karkaria et al., Nat Commun, 2021
Karkaria et al., PLOS Comp Biol, 2022

Inputs

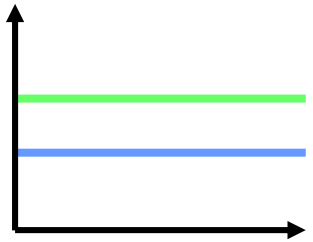
Model space



Parameter space

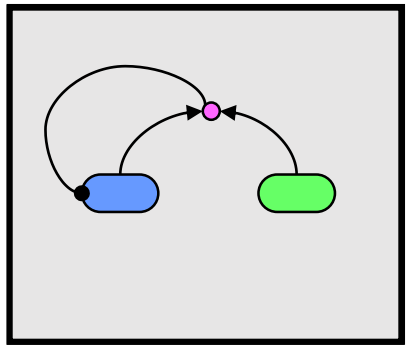


Desired behaviour

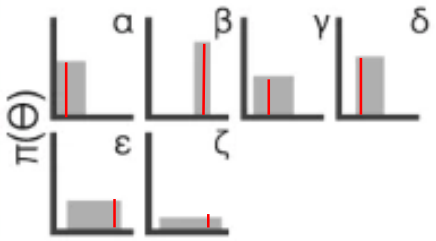


ABC

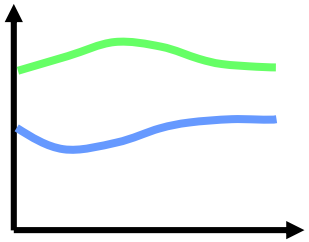
Sample a model



Sample parameters

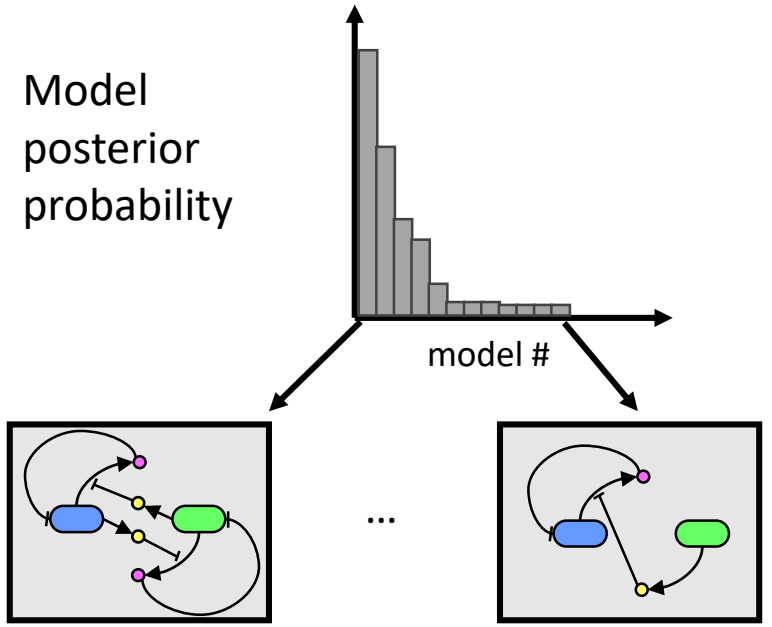


Simulate model and compare to desired behaviour



Output

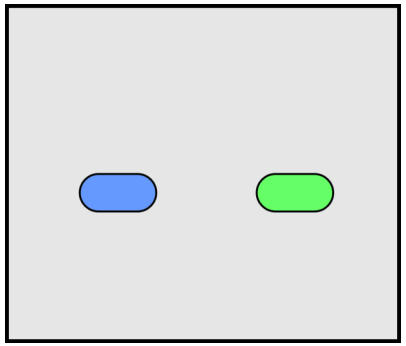
Model posterior probability



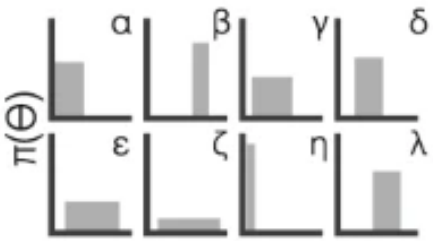
Karkaria et al., Nat Commun, 2021
Karkaria et al., PLOS Comp Biol, 2022

Inputs

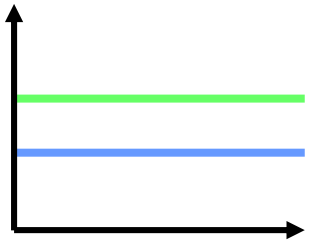
Model space



Parameter space

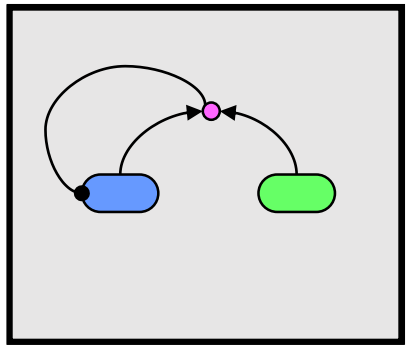


Desired behaviour

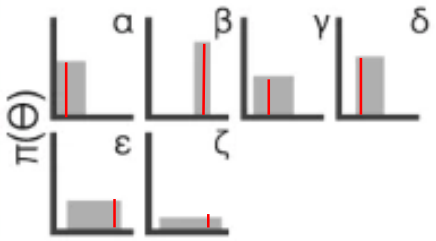


ABC

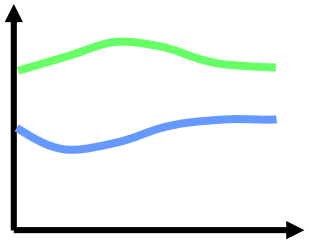
Sample a model



Sample parameters

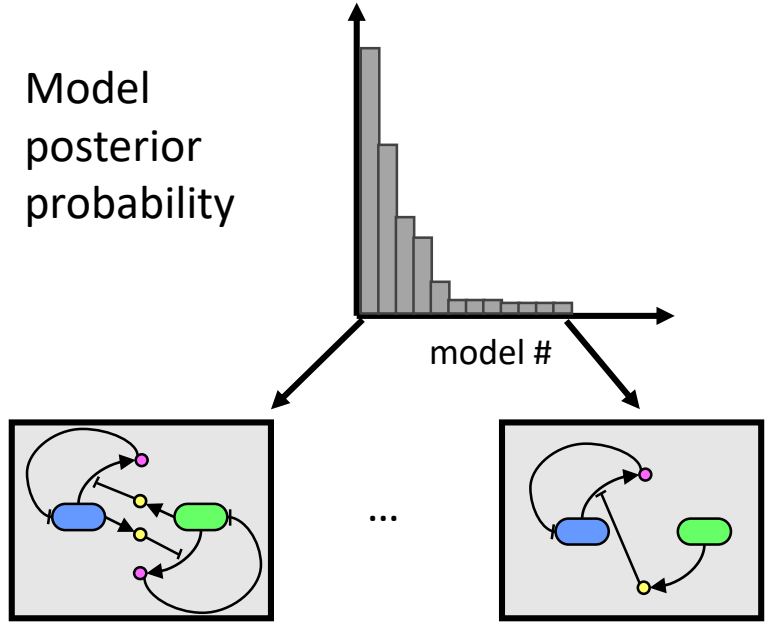


Simulate model and compare to desired behaviour

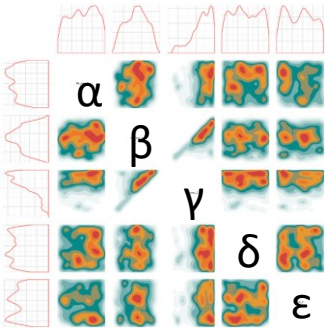


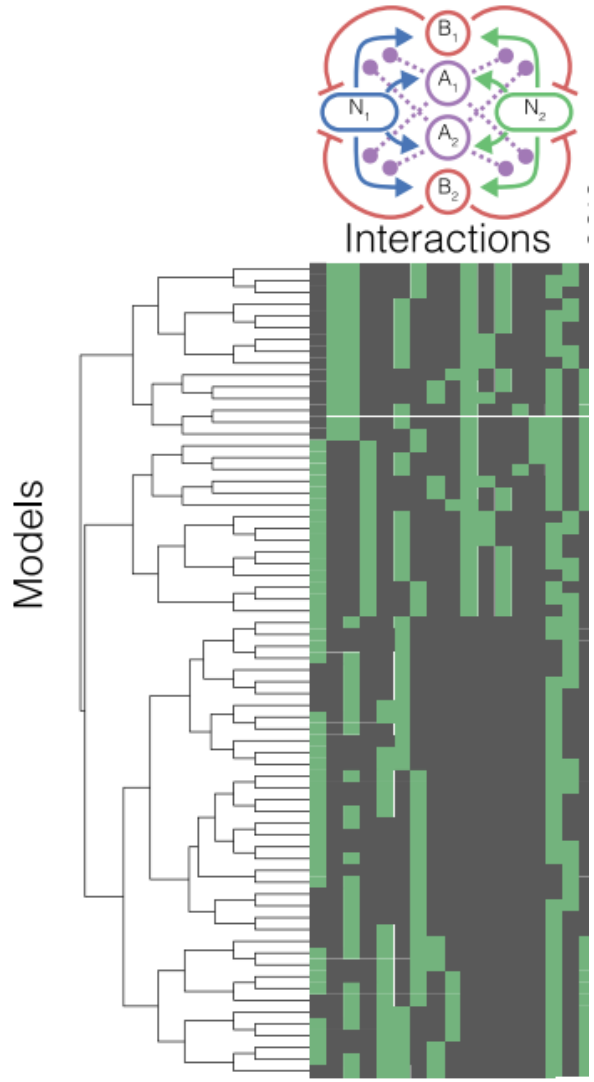
Output

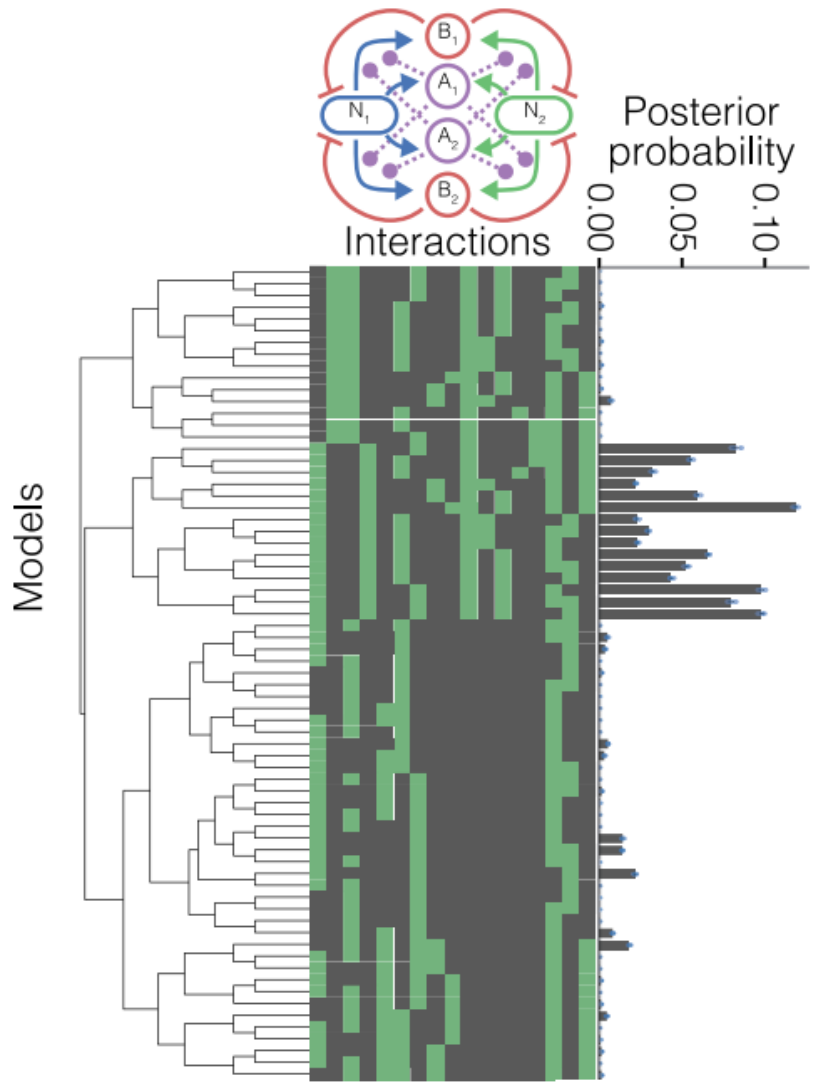
Model posterior probability

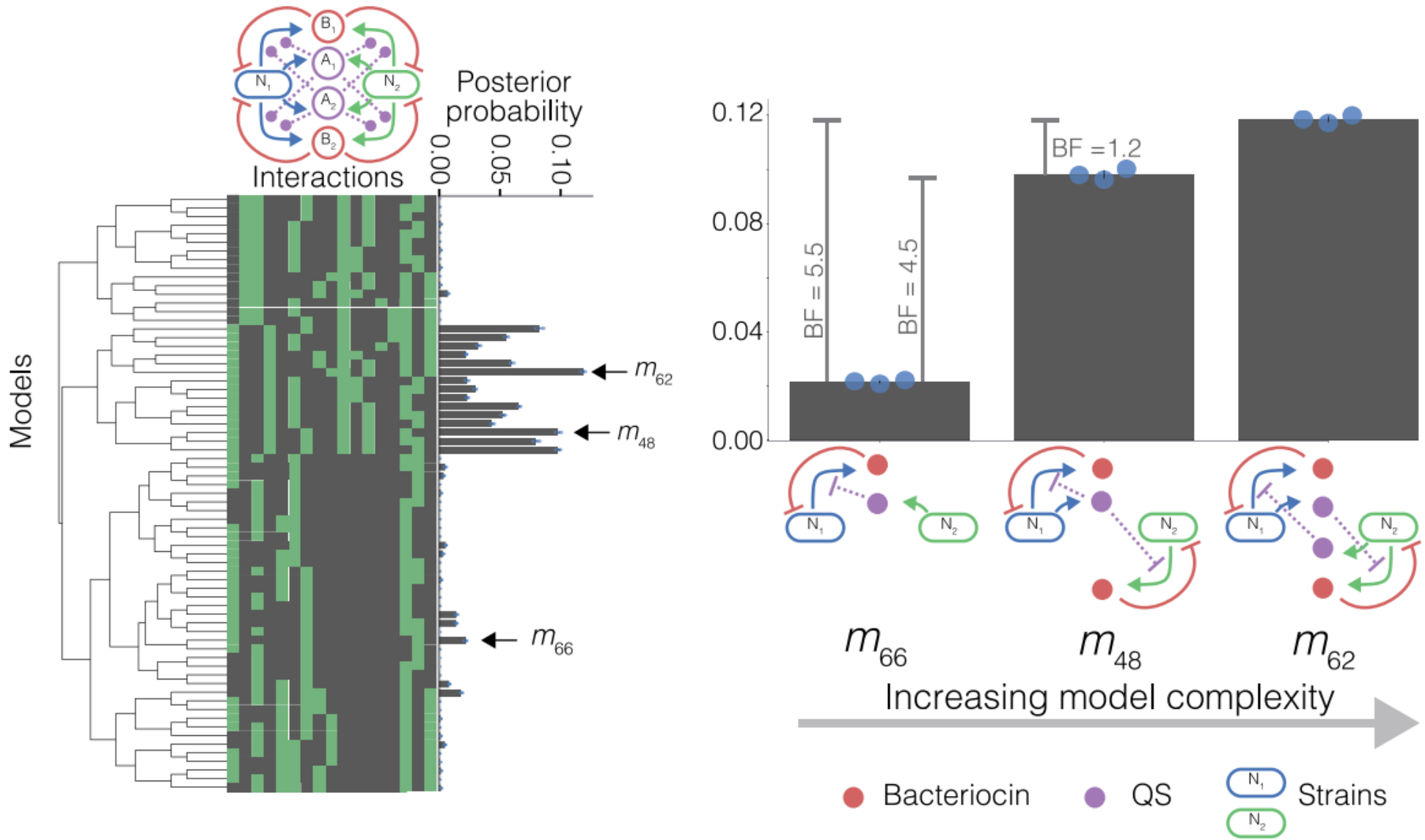


Parameter posterior probability

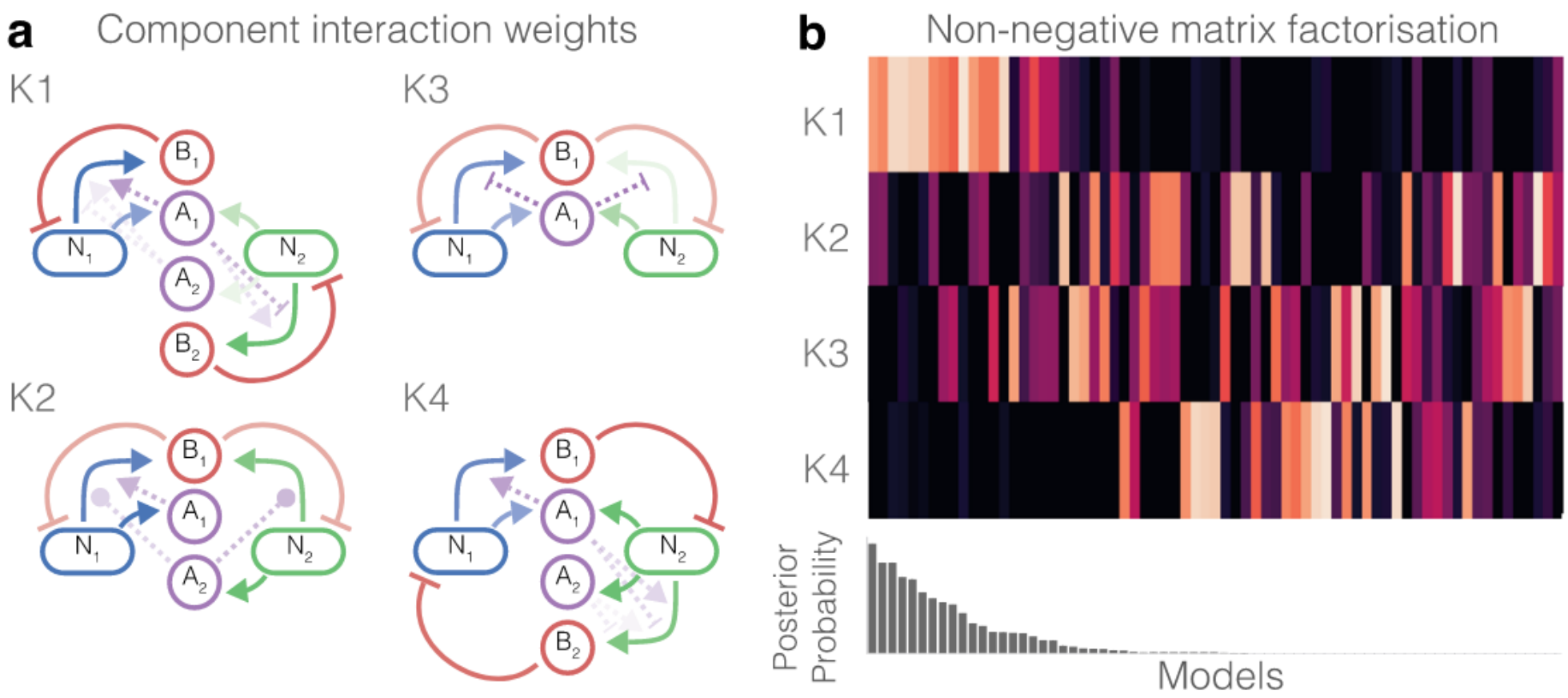



B1
A1
N1
N2
A2
B2
Interactions
Models

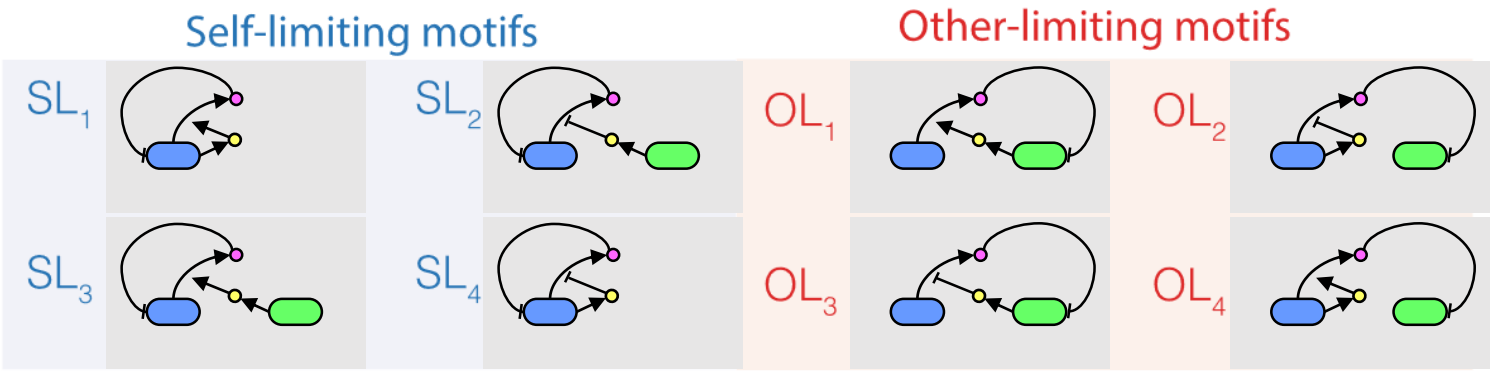

B1
A1
N1
N2
A2
B2
Interactions
Posterior probability
0.00
0.05
0.10
Models

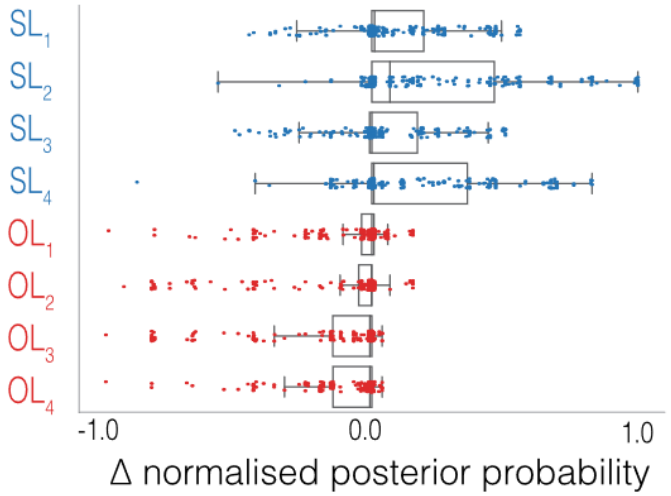

B1
A1
N1
N2
A2
B2
Interactions
Posterior probability
0.00
0.05
0.10
Models
m62
m48
m66
0.12
0.08
0.04
0.00
BF = 5.5
BF = 4.5
BF =1.2
m66
m48
m62
Increasing model complexity
Bacteriocin
QS
Strains

Are there general rules that I can follow?

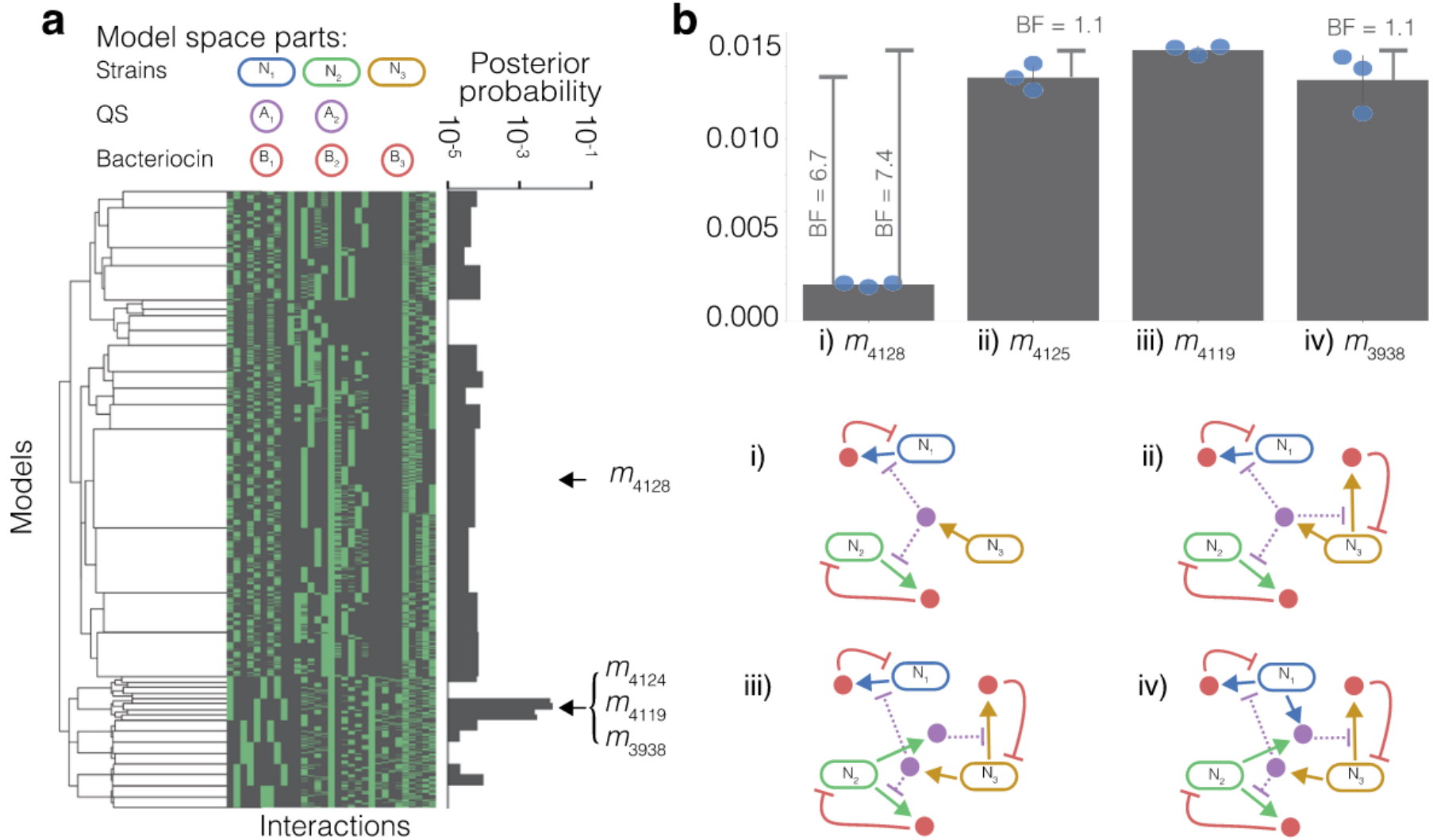


Are there general rules that I can follow?





Karkaria et al., Nat Commun, 2021


a
Model space parts:
Strains
QS
Bacteriocin
Posterior probability
Models
Interactions
m_{4128}
m_{4124}
m_{4119}
m_{3938}
b
BF = 6.7
BF = 7.4
BF = 1.1
BF = 1.1
i) m_{4128}
ii) m_{4125}
iii) m_{4119}
iv) m_{3938}

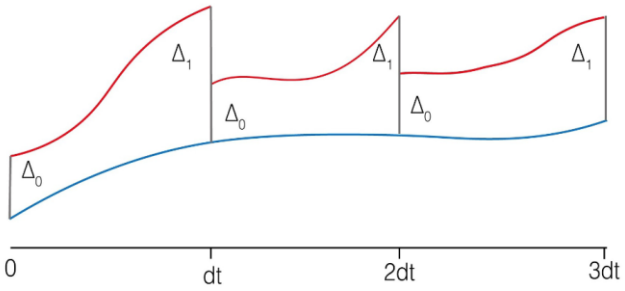
Using ABC to find rare behaviours (chaos)

Generalised Lotka-Volterra

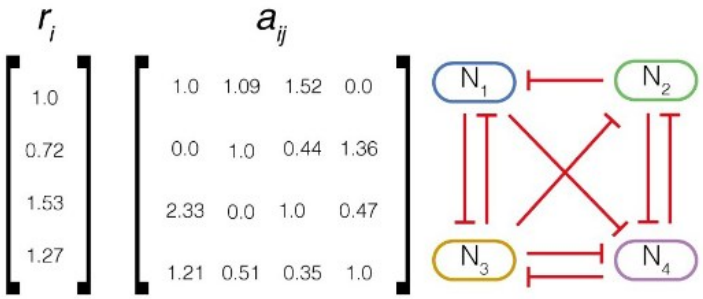
$$\frac{dX_i}{dt} = X_i\left(\mu_i + \sum_j \alpha_{i,j} X_j\right)$$

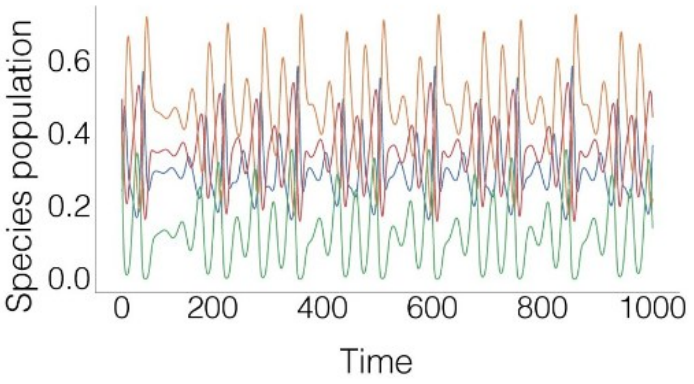
Distance function:

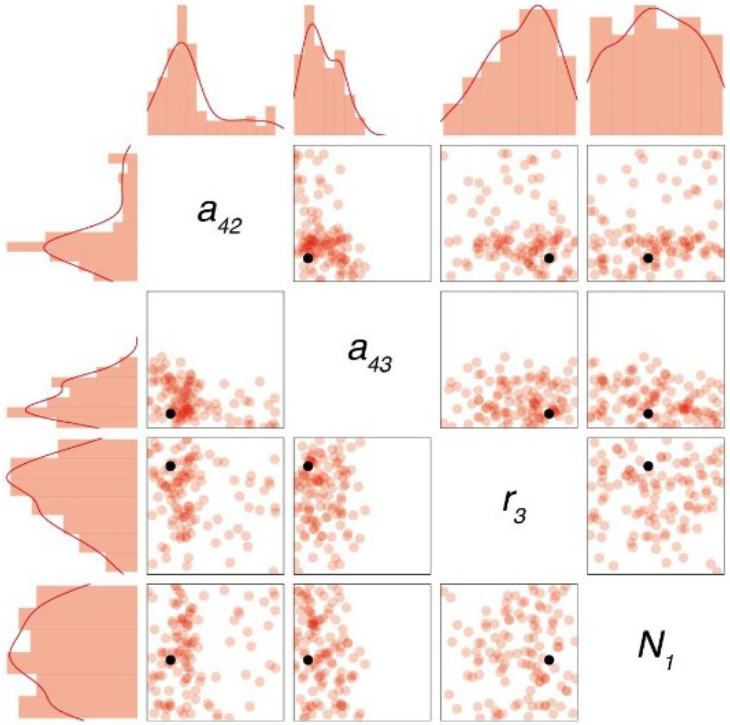
$d_1 = {}^{1}/_{(1+\lambda_1)}$ for $\lambda_1 > 0$



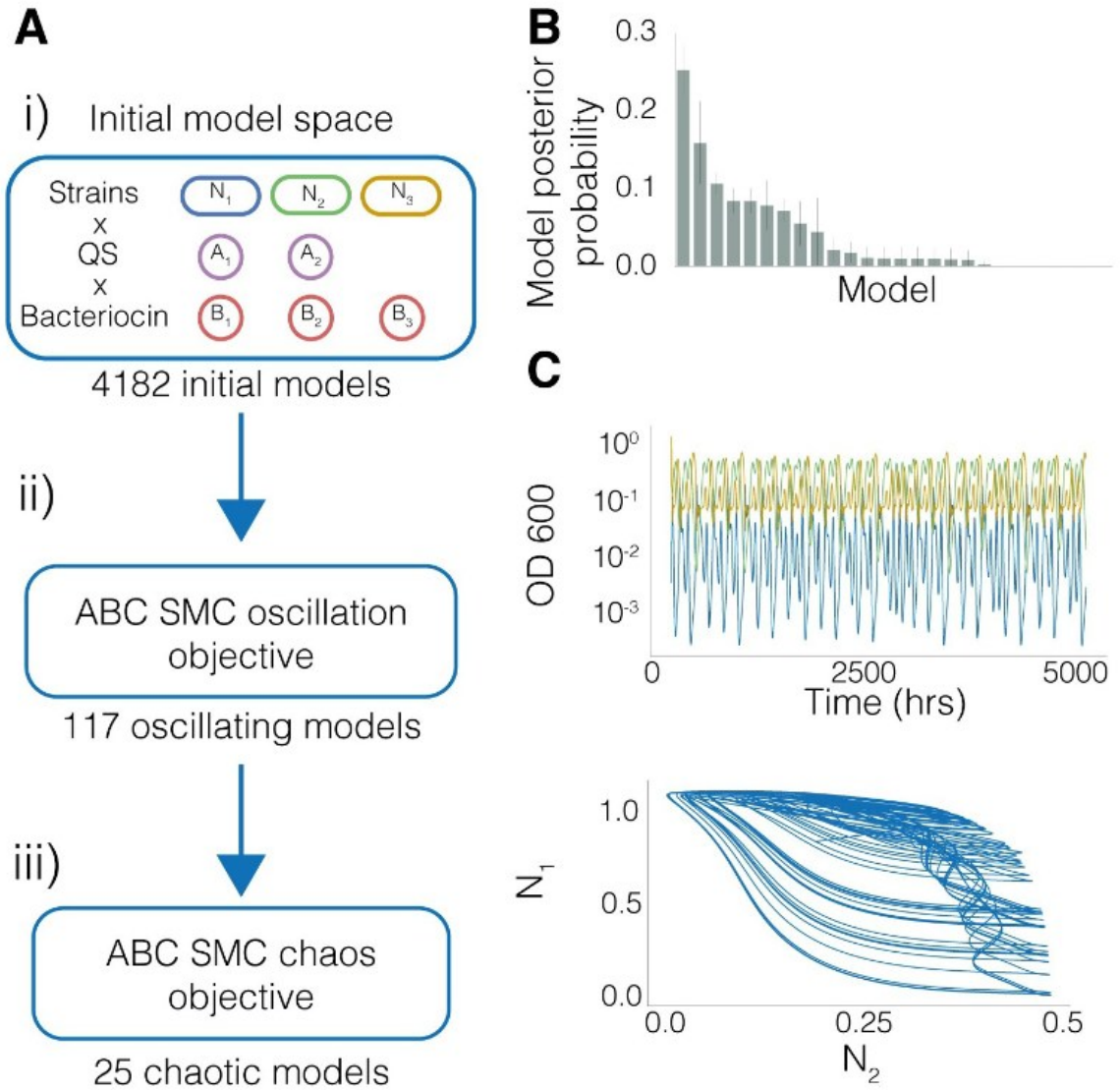
Sprott JC. Chaos and Time-Series Analysis; 2003



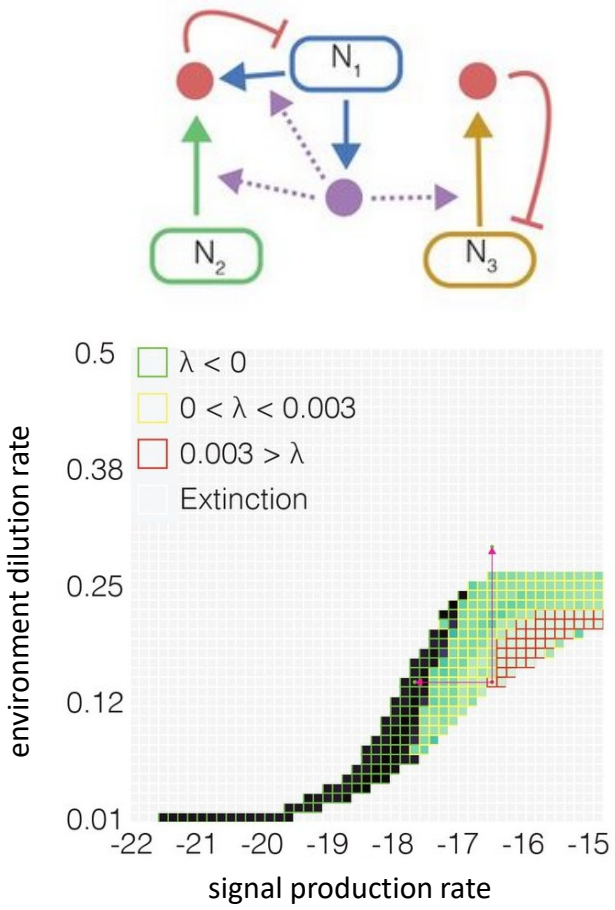


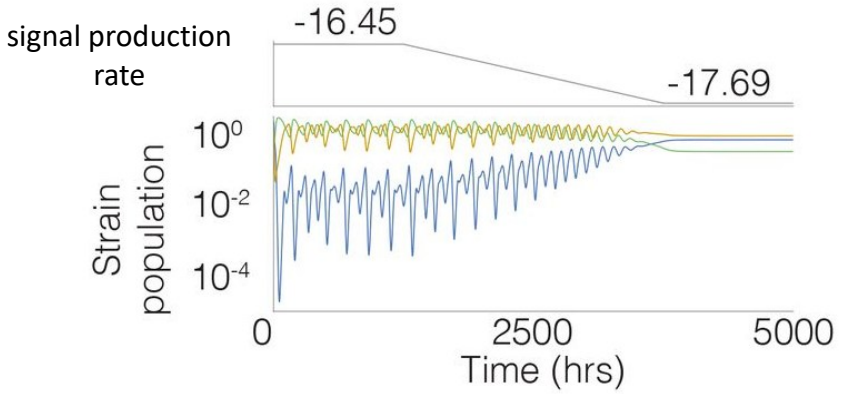


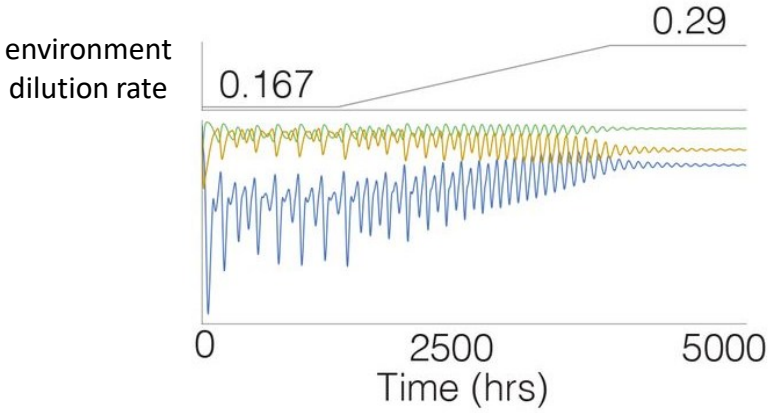
Oscillations as a route to chaos



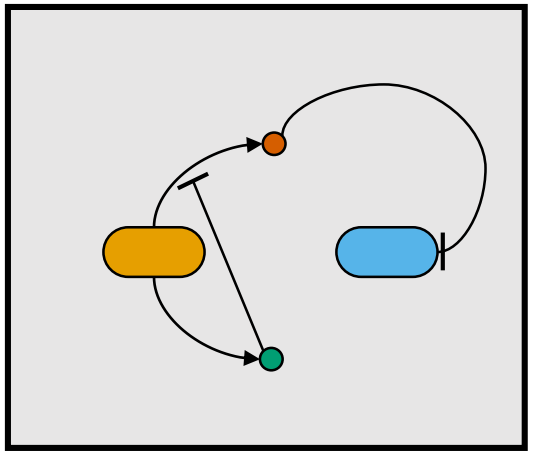
Moving from chaos to stability

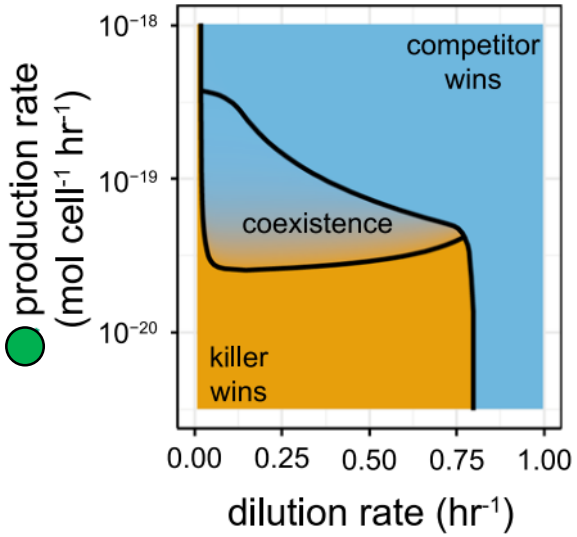


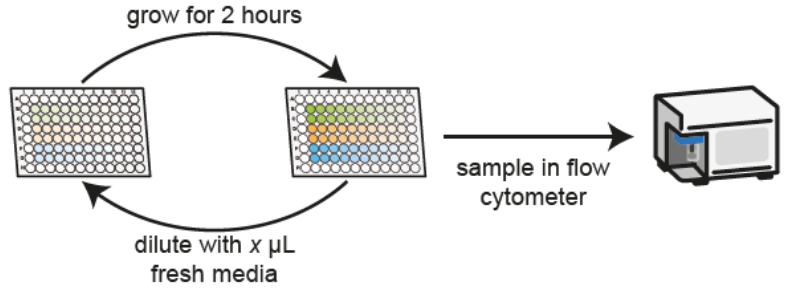


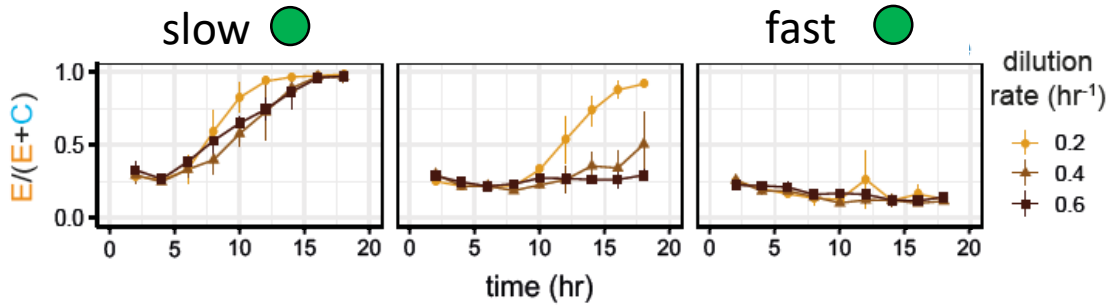


So that we can rapidly build and test communities

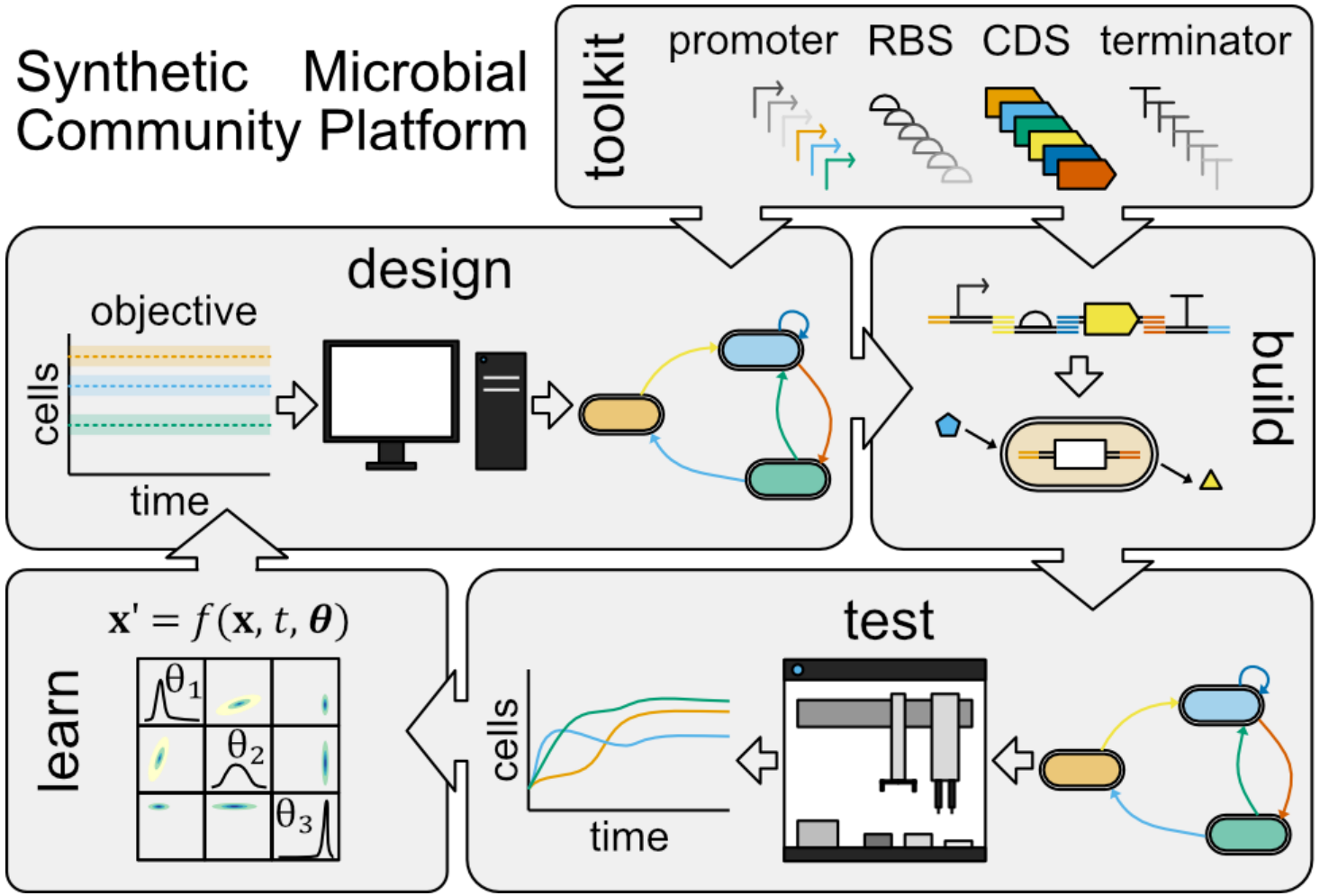








Fedorec et al., Nat Commun, 2021



I’m hiring postdocs for this work!

Acknowledgements

Fedorec Group
Casey Chen
Alice Cook
Louie Destouches

Chengtai Yin
Donny Chang
Miranda Shou
Mod Jantorn
Griffin Keidaisch
Paul Goodman

Chris Barnes
Behzad Karkaria
Neythen Treloar

Biochemical Engineering, UCL
Michael Sulu

Mathematics, UCL
Angelika Manhart
Alexey Zaikin

University of Waterloo
Brian Ingalls

UKRI Biotechnology and Biological Sciences Research Council

UKRI Engineering and Physical Sciences Research Council

erc
European Research Council
Established by the European Commission